# Contents



Exercises are from Stephen Abbott's *Understanding Analysis*, Second Edition.

## Problem 1: Functional limits and function composition

(Exercise 4.3.4)

### Proof Idea

This question is all about the difference between continuity and functional limits, namely, what's going on exactly *at c*. For functional limits, anything could be happening there. But for continuous functions, $\lim_{x \to c} f(x)$ must actually equal $f(c)$.

We're looking for a counterexample to a statement we *know* is true for continuous functions,. So we've got to mess around with the value at $c$ and make it different from the limit.

### (a)

Let

$$g(x) = \begin{cases} 0 \text{ if } x = 0 \\ 1 \text{ otherwise} \end{cases}$$

$$\text{and}$$

$$f(x) = 0.$$

Then $\lim_{x \to 0} f(x) = 0$ and $\lim_{x \to 0} g(x) = 1$, but $\lim_{x \to 0} g(f(x)) = 0$ since $g(f(x)) = 0$ for all values of $x$.

### (b)

Suppose $f$ and $g$ are continuous. Since the functions are defined on all of $\mathbb{R}$, every point $c \in \mathbb{R}$ is a limit point of the domain. Therefore

$$\lim_{x \to p} f(x) = f(p) = q$$

$$\text{and}$$

$$\lim_{x \to q} g(x) = g(q) = r.$$

Also, $g \circ f$ is continuous, as proved in the textbook and also in a previous writeup, so

$$\lim_{x \to p} g(f(x)) = g(f(p)) = g(q) = r.$$

### (c)

As the counterexample shows, the result in (a) does not hold if only $f$ is continuous. But continuity of $g$ is enough:

**Theorem 1**: If $g : \mathbb{R} \to \mathbb{R}$ is continuous and $\lim_{x \to p} f(x) = q$ and $\lim_{x \to q} g(x) = r$, then

$$\lim_{x \to p} g(f(x)) = r.$$

*Proof*: Since $f$ and $g$ are defined on all of $\mathbb{R}$, every point in $\mathbb{R}$ is a limit point of their domain. By continuity of $g$ at $q$, $\lim_{x \to q} g(x) = g(q) = r$.

Let $B_\varepsilon(r)$ be any neighborhood of $r = g(q)$. Then by continuity of $g$, there's a neighborhood $B_\eta(q)$ of $q$ such that $g\big(B_\eta(q)\big) \subseteq B_\varepsilon(r)$. (Since $g$ is continuous, the *entire* neighborhood gets mapped inside the $\varepsilon$-ball around $r$, including $x = q$.)

And because $\lim_{x \to p} f(x) = q$, corresponding to the neighborhood $B_\eta(q)$, there's a neighborhood $B_\delta(p)$ of $p$ such that for every $x \in B_\delta(p)$ such that $x \neq p$, we have $f(x) \in B_\eta(q)$. But since $g$ maps every point in $B_\eta(q)$ inside $B_\varepsilon(r)$, we have

$$g(f(x)) \in B_{\varepsilon(r)} \text{ whenever } x \in B_\delta(p), x \neq p.$$

In other words,

$$\lim_{x \to p} g(f(x)) = r.$$

■

**Remark 0.1**: This proof goes through because $g$'s continuity means that *every* point in a neighborhood around its input gets mapped into the target ball. That is where the proof would fail if all we knew was that $\lim_{x \to q} g(x) = r$.

## Problem 2: Contraction Mapping Theorem

(Exercise 4.3.11)

#### Proof idea

Speaking informally and intuitively, every application of the function $f$ only moves us a factor of $c$ away from where we were before. So we should be able to relate the iteration to terms in a geometric series.

To make this precise, apply the triangle inequality to the thing we need to bound for the Cauchy Criterion.

Apply the sequential characterization of continuity to $(y_n)$ to obtain that it's a fixed point.

### Proofs

Let $f : \mathbb{R} \to \mathbb{R}$ and $c$ a constant such that $0 < c < 1$ and

$$|f(x) - f(y)| \leq c|x - y|$$

for all $x, y \in \mathbb{R}$.

### (a)

**Theorem 2**: $f$ is continuous on $\mathbb{R}$.

*Proof*: Let $a \in \mathbb{R}$ be any point. Then $a$ is in the domain of $f$ since $f$ is defined on all of $\mathbb{R}$. Let $\varepsilon > 0$ be given and set $\delta = \frac{\varepsilon}{c}$. Then, whenever

$$|x - a| < \delta$$

we have

$$|f(x) - f(a)| \leq c|x - a| < c\delta = \varepsilon.$$

So $f$ is continuous at $a$. Since $a$ was arbitrary, $f$ is continuous on all of $\mathbb{R}$. ■

### (b)

**Theorem 3**: Let $y_1 \in \mathbb{R}$ be any point and define the sequence $(y_n)$ by

$$y_{n+1} = f(y_n)$$

for $n \geq 1$. Then $(y_n)$ is Cauchy.

*Proof*: Define

$$z_n = |y_n - y_{n-1}|$$

for $n > 1$.

Then by the given properties of the function $f$ and the definition of the sequence $(y_n)$,

$$
\begin{aligned}
z_n &= |y_n - y_{n-1}| \\
&= |f(y_{n-1}) - f(y_{n-2})| \\
&\leq c|y_{n-1} - y_{n-2}| \\
&= cz_{n-1}
\end{aligned}
$$

for $n > 2$.

So, when $n$ is large enough that all the $z_i$'s below are defined, we have

$$z_n \leq cz_{n-1} \leq c^2 z_{n-2} \leq \dots \leq c^m z_{n-m} \leq \dots \tag{1}$$

Let $\varepsilon > 0$ be given. To show that the sequence $(y_n)$ is Cauchy, we must find some $N \in \mathbb{N}$ such that whenever $m > n \geq N$ we have $|y_m - y_n| < \varepsilon$. To bound this quantity we add and subtract all $y_i$'s between $m$ and $n$, then apply the triangle inequality:

$$
\begin{aligned}
&|y_m - y_n| \\
&= |y_m - y_{m-1} + y_{m-1} - y_{m-2} + y_{m-2} - \dots + y_{n+1} - y_n| \\
&\leq |y_m - y_{m-1}| + |y_{m-1} - y_{m-2}| + \dots + |y_{n+1} - y_n| \\
&= z_m + z_{m-1} + \dots + z_{n+1}.
\end{aligned}
$$

Here we apply Equation 1 to bound each $z_i$ by a multiple of $z_{n+1}$:

$$
\begin{aligned}
&z_m + z_{m-1} + \dots + z_{n+1} \\
&\leq c^{m-n-1} z_{n+1} + c^{m-n-2} z_{n+1} + \dots c z_{n+1} + z_{n+1} \\
&= z_{n+1} \sum_{i=0}^{m-n-1} c^i \\
&< z_{n+1} \frac{1}{1-c}
\end{aligned}
$$

where the last inequality follows because sum in the previous step is a partial sum of the geometric series $\sum_{i=0}^{\infty} c^i$, and every term in this series is positive since $0 < c < 1$, and the series converges because $|c| < 1$.

Now, the quantity $z_{n+1}$ itself may be bounded by another application of Equation 1:

$$z_{n+1} \leq c^{n-1} z_2.$$

So

$$|y_m - y_n| < \frac{c^{n-1}}{1-c} z_2.$$

Since $0 < c < 1$, we may choose $N$ so that

$$\frac{c^{N-1}}{1-c} z_2 < \varepsilon;$$

such an $N$ is guaranteed to exist because the sequence $(c^n)$ converges to 0. (Why? I reproduce the proof in the book as review for myself: the sequence converges to some $l$ by the Monotone Convergence Theorem. $(c^{2n})$ is a subsequence that must converge to the same $l$. But $c^{2n} = c^n c^n$ so by the Algebraic Limit Theorem, $l^2 = l$, and $l$ cannot be 1.) ∎

Hence we may let $y = \lim y_n$.

## (c)

**Theorem 4**: Let $y = \lim y_n$. Then $y$ is a fixed point of $f$, that is,

$$y = f(y)$$

*Proof*: We have proved that $f$ is continuous. By the Sequential Characterization of Continuity, since $(y_n) \to y$, we must have

$$(f(y_n)) \to f(y).$$

But the sequence $(f(y_n))$ is $(f(y_1), f(y_2), \ldots)$, that is, the same as $(y_n)$ without the first term. So it is a subsequence, and must therefore converge to the same limit. So

$$(f(y_n)) \to y.$$

Limits of sequences are unique, so $y = f(y)$. ■

**Theorem 5**: There is a unique $y \in \mathbb{R}$ such that

$$y = f(y)$$

*Proof*: The preceding theorems demonstrate the existence of at least one $y$ such that $y = f(y)$.

Suppose for the sake of contradiction that there existed distinct $x, y \in \mathbb{R}$ satisfying $x = f(x)$ and $y = f(y)$. By the defining property of $f$,

$$|f(x) - f(y)| \leq c|x - y|.$$

Substituting $x = f(x)$ and $y = f(y)$,

$$|x - y| \leq c|x - y|.$$

Since $x \neq y$, the quantity $|x - y|$ is nonzero and so we may divide by it to obtain

$$c \geq 1,$$

a contradiction of $0 < c < 1$. ■

## (d)

**Theorem 6**: If $x$ is any arbitrary point in $\mathbb{R}$, then the sequence $(x, f(x), f(f(x)), \ldots)$ converges to $y$ defined in part (b).

*Proof*: Let $(x_n)$ be the sequence $(x, f(x), f(f(x)), \ldots)$. Since $y_1$ in part (b) was arbitrary, the same proofs apply to $(x_n)$, so $(x_n)$ converges. Let $l = \lim x_n$. Then by part (c), $l$ must satisfy $l = f(l)$. By the previous result, $l = y$. ■

## Problem 3: Uniform Continuity of $1/x^2$

(Exercise 4.4.3)

**Theorem 7**: $f(x) = 1/x^2$ is uniformly continuous on $[1, \infty)$.

*Proof*: Let $\varepsilon > 0$ be given and set $\delta = \min\left(\frac{1}{2}, \frac{\varepsilon}{10}\right)$.

Let $x, c \in [1, \infty)$ be any points in the given domain with $|x - c| < \delta$. Then

$$c \geq 1$$
so
$$0 < \frac{1}{c} \leq 1.$$

We will need a bound for $|x + c|$ soon. Write $x + c = x + c - c + c = 2c + x - c$ and apply the triangle inequality to get

$$|x + c| \leq 2|c| + |x - c| = 2c + \delta$$

so

$$\frac{|x + c|}{|c|} \leq 2 + \frac{\delta}{c}$$

Also, since $\delta \leq \frac{1}{2}$ and $c \geq 1$,

$$x > c - \delta \geq \frac{1}{2}$$
so
$$\frac{1}{x} < 2.$$

So

$$\left|\frac{1}{x^2} - \frac{1}{c^2}\right| = \left|\frac{c^2 - x^2}{c^2 x^2}\right| = \left|\frac{c - x}{cx}\right|\left|\frac{c + x}{cx}\right|.$$

Now we have bounds for every factor in this last expression:

$$\begin{aligned} |c - x| &< \delta, \\ \left|\frac{1}{c}\right| &\leq 1, \\ \left|\frac{1}{x}\right| &\leq 2 \text{ (which appears twice)}, \\ \frac{|c + x|}{|c|} &\leq 2 + \frac{\delta}{c} \leq 2 + \delta, \end{aligned}$$

which we multiply together to obtain

$$\left|\frac{1}{x^2} - \frac{1}{c^2}\right| < \delta \cdot 2 \cdot 2 \cdot (2 + \delta) = 4\delta(\delta + 2).$$

Since $\delta \leq \frac{1}{2}$ we have $\delta + 2 \leq \frac{5}{2}$, so

$$4\delta(\delta + 2) \leq 10\delta \leq \varepsilon$$

as required. ∎

**Theorem 8**: $f(x) = 1/x^2$ is not uniformly continuous on $(0, 1]$.

*Proof*: Define

$$x_n = \cos\left(\frac{n}{n+1}\frac{\pi}{2}\right)$$
and
$$y_n = \cot\left(\frac{n}{n+1}\frac{\pi}{2}\right).$$

Let

$$\theta_n = \frac{n}{n+1}\frac{\pi}{2}.$$

Then for every $n \in \mathbb{N}$, since $\theta_n \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right)$, both $\cos\theta_n$ and $\cot\theta_n$ are positive and at most 1. Hence, both $x_n$ and $y_n$ are in $(0, 1]$.

Also,

$$|x_n - y_n| = |\cos\theta_n - \cot\theta_n| = \left|\frac{\cos\theta_n(\sin\theta_n - 1)}{\sin\theta_n}\right|$$

As $n \to \infty$,

$$\theta_n \to \frac{\pi}{2}$$

so by continuity of the sine and cosine functions and the sequential criterion of continuity,

$$\cos\theta_n \to 0,$$
$$\sin\theta_n \to 1,$$
$$\sin\theta_n \neq 0$$

so $|x_n - y_n| \to 0$ by the Algebraic Limit Theorem and continuity of the absolute value function.

But

$$|f(x_n) - f(y_n)| = \left|\frac{1}{x_n^2} - \frac{1}{y_n^2}\right| = \left|\sec^2\theta_n - \tan^2\theta_n\right| = 1$$

for every $n$.

Put $\varepsilon_0 = 1$ in the Sequential Criterion for Absence of Uniform Continuity. Then, since $|x_n - y_n| \to 0$ but $|f(x_n) - f(y_n)| = 1 \geq \varepsilon_0$, the function $f(x) = 1/x^2$ fails to be uniformly continuous on $(0, 1]$. ∎

**Remark 0.2**: This example is maybe a bit cheap since one could argue that all the hard work is happening in the definitions of the trigonometric functions and the proofs of their properties.

**Remark 0.3**: It feels like we should be able to figure out some properties of $1/x^2$ and the two domains, and prove theorems relating those properties to uniform continuity. Maybe something to do with being monotone? Or bounded vs unbounded?

On $[1, \infty)$ the function is bounded and monotone (and continuous). Maybe that's enough to prove uniform continuity.

On $(0, 1]$ the function is unbounded. Unboundedness is not enough to make a function not uniformly continuous, though. For example the identity function is unbounded, monotone, continuous and uniformly continuous on $[0, \infty)$.

**Remark 0.4**: Actually, maybe we can use the Continuous Extension Theorem from the next problem to say something like: $1/x^2$ is unbounded in every neighborhood of $x = 0$, therefore it can have no continuous extension on any interval containing 0, therefore it is not uniformly continuous on $(0, 1)$, therefore it is not uniformly continuous on $(0, 1]$.

## Problem 4: Continuous Extension Theorem

(Exercise 4.4.13)

### (a)

**Theorem 9**: Suppose that $f : A \to \mathbb{R}$ is uniformly continuous and $(x_n) \subseteq A$ is a Cauchy sequence. Then $(f(x_n))$ is Cauchy.

*Proof*: Let $\varepsilon > 0$ be given. By uniform continuity of $f$ on $A$, there exists some $\delta > 0$ such that for all $x, y \in A$, whenever $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$.

Since $(x_n)$ is Cauchy, there exists some $N \in \mathbb{N}$ such that whenever $m > n \geq N$, we have $|x_m - x_n| < \delta$.

Therefore whenever $m > n \geq N$, since $|x_m - x_n| < \delta$, we have $|f(x_n) - f(x_m)| < \varepsilon$, showing that $(f(x_n))$ is Cauchy. ∎

### (b)

**Theorem 10**: Let $g$ be a continuous function on the open interval $(a, b)$. Then $g$ is uniformly continuous on $(a, b)$ iff it is possible to define values $g(a)$ and $g(b)$ at the endpoints so that the extended function $g$ is continuous on $[a, b]$.

*Proof*: ($\Rightarrow$) We may assume $a \neq b$, otherwise the set $(a, b)$ is empty, $g$ takes no values there and there is nothing to prove.

Suppose that $g$ is uniformly continuous on $(a, b)$. Then define the sequences

$$
\begin{aligned}
x_n &= a + \frac{b - a}{2n}, \\
y_n &= b - \frac{b - a}{2n}
\end{aligned}
$$

so that $(x_n) \to a$ and $(y_n) \to b$, and both sequences have all their terms in $(a, b)$. Since the sequences converge, they are Cauchy. Then by the previous result, $g(x_n)$ and $g(y_n)$ are also Cauchy, and therefore converge. Define $g(a) = \lim g(x_n)$ and $g(b) = \lim g(y_n)$.

Define the function

$$
h(x) = \begin{cases} g(x) \text{ if } x \in (a, b) \\ g(a) \text{ if } x = a \\ g(b) \text{ if } x = b \end{cases}.
$$

Let $(z_n) \subseteq [a, b]$ be any sequence converging to $a$ with $z_n \neq a$. Then after finitely many terms we must also have $z_n \neq b$, since $|b - a| > 0$ and all terms of $(z_n)$ must eventually be less than that distance from $a$. So after dropping that finite prefix and replacing $(z_n)$ by its tail, $(z_n) \subseteq (a, b)$.

Then $(x_n - z_n) \to 0$ by the Algebraic Limit Theorem. Also, since $(z_n)$ is convergent, it is Cauchy, so by the previous theorem $(g(z_n))$ is Cauchy. Therefore it converges to some limit, call it $l$. Therefore $(g(x_n) - g(z_n)) \to g(a) - l$ by the Algebraic Limit Theorem.

We claim that $(g(x_n) - g(z_n)) \to 0$: let $\varepsilon > 0$ be given. Then by uniform continuity of $g$ on $(a, b)$, there exists some $\delta > 0$ such that $|x_n - z_n| < \delta \Rightarrow |g(x_n) - g(z_n)| < \varepsilon$. By convergence of $(x_n - z_n)$ to 0, there exists some $N \in \mathbb{N}$ such that when $n \geq N$, $|x_n - z_n| < \delta$. Therefore, for $n \geq N$ we have $|g(x_n) - g(z_n)| < \varepsilon$, proving convergence to 0.

Since limits of sequences are unique, we have $g(a) - l = 0$ so $l = g(a)$.

Therefore, by the Sequential Criterion for Functional Limits, $\lim_{x \to a} h(x) = g(a)$. But then, since $a$ is a limit point of the domain of $h$, $[a, b]$, by Theorem 4.3.2 (Characterizations of Continuity) part (iv), we have $h(x)$ continuous at $a$.

(This argument shows both that $g(a)$ is well-defined, and that $h(x)$ is continuous at $a$.)

By a symmetric argument, $h(x)$ is continuous at $b$. Also, $h(x)$ is continuous at every point in $(a, b)$, i.e., adding endpoints doesn't matter there: since that region is open, for any $c \in (a, b)$ there's a neighborhood of $c$ contained entirely in $(a, b)$ where $h$ and $g$ agree, so by continuity of $g$ on $(a, b)$ and using the $\varepsilon - \delta$ definition of continuity, $h$ is continuous at every $c \in (a, b)$.

Therefore the extension is continuous on $[a, b]$.

$(\Leftarrow)$

Suppose that it is possible to define values $g(a)$ and $g(b)$ at the endpoints so that the extended function $g$ is continuous on $[a, b]$. Since the closed interval is compact, the extended $g$ is uniformly continuous on $[a, b]$.

By the definition of uniform continuity, for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all $x, y \in [a, b]$, $|x - y| < \delta$ implies $|g(x) - g(y)| < \varepsilon$. Since the condition holds for all $x, y \in [a, b]$, it also holds for all $x, y \in (a, b)$ or any other subset of $[a, b]$. So $g$ is uniformly continuous on $(a, b)$ as well. ∎

**Remark 0.5**: I do a weird tricky thing here by using sequences that can't actually take on values at the endpoints, to prove continuity at the endpoints. To do this I had to go through the Functional Limit definition. It seems bad. I think the proof might be correct but I wonder if there's a clearer way to see it.

## Problem 5: Intermediate Value Theorem, Two Ways

(Exercise 4.5.5)

### (a) Using the Axiom of Completeness

**Lemma 1**: Let $f : [a, b] \to \mathbb{R}$ be continuous and $f(a) < 0 < f(b)$. Then $f(c) = 0$ for some $c \in (a, b)$.

*Proof*: Define

$$K = \{x \in [a, b] : f(x) \leq 0\}.$$

Then $a \in K$ since $f(a) < 0$, and $K$ is bounded above by $b$ since $K \subseteq [a, b]$ by construction. By the Axiom of Completeness, $K$ has a least upper bound. Let $c = \sup K$. Then $c \leq b$ since $b$ is an upper bound and $c$ the *least* upper bound. So $f$ is continuous at $c$.

By the ordering axioms, either $f(c) < 0$ or $f(c) > 0$ or $f(c) = 0$.

If $f(c) > 0$ let $\varepsilon = f(c) > 0$. Then by continuity of $f$ at $c$, there exists some $\delta > 0$ such that whenever $|x - c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$. So

$$-\varepsilon < f(x) - f(c) < \varepsilon$$

so, since $\varepsilon = f(c)$,

$$f(x) > 0$$

for every $x \in (c - \delta, c + \delta)$. But by the defining property of least upper bound, there must be some $\alpha \in K$ satisfying $c - \delta < \alpha < c$, a contradiction.

If, on the other hand, $f(c) < 0$ then let $\varepsilon = -f(c) > 0$. By continuity of $f$ at $c$ there exists some $\delta > 0$ such that whenever $|x - c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$. So

$$-\varepsilon < f(x) - f(c) < \varepsilon$$

so, since $\varepsilon = -f(c)$,

$$f(x) < 0$$

for every $x \in (c - \delta, c + \delta)$. So $f\left(c + \frac{\delta}{2}\right) < 0$, so $c + \frac{\delta}{2} \in K$, so $c$ is not an upper bound for $K$.

The only remaining possibility is that $f(c) = 0$.

We now show that $c \in (a, b)$. Since $a \in K$ and $c = \sup K$ we have $c \geq a$. We previously showed that $c \leq b$. But $f(c) = 0$ while $f(a) < 0 < f(b)$, so $c \neq a$ and $c \neq b$. Therefore $c \in (a, b)$. ■

### (b) Using Nested Interval Property

**Lemma 2**: Let $f : [a, b] \to \mathbb{R}$ be continuous and $f(a) < 0 < f(b)$. Then $f(c) = 0$ for some $c \in (a, b)$.

*Proof*: Let $I_0 = [a, b]$ and set $z_0 = \frac{a+b}{2}$, that is, its midpoint. If $f(z_0) \geq 0$ then set $a_1 = a$ and $b_1 = z_0$. Otherwise it must be that $f(z_0) < 0$, in which case set $a_1 = z_0$ and $b_1 = b$.

Either way, the interval $I_1 = [a_1, b_1]$ has the property that $f(a_1) < 0$ and $f(b_1) \geq 0$ and $\text{len}(I_1) = \text{len}(I_0)/2$ and $I_0 \supset I_1$.

Continuing in this way, form the sequence of intervals $I_0 \supset I_1 \supset I_2 \supset \ldots \supset I_n \supset \ldots$. Then $\text{len}(I_k) = (b-a)/2^k$. By the Nested Interval Property, there exists a real number $c$ that's in every interval $I_k$.

$f$ is continuous at $c$ since $c \in I_0 = [a, b]$. Then by the ordering axioms, either $f(c) < 0$ or $f(c) > 0$ or $f(c) = 0$.

If $f(c) > 0$ then let $\varepsilon = f(c) > 0$ in the definition of continuity. Then there must exist some $\delta > 0$ such that $f(x) > 0$ for every $x$ in that neighborhood. But, by the Archimedean Principle, there's some $k \in \mathbb{N}$ such that the length of $I_k$ is smaller than $\delta$. Then, since $c \in I_k$ and $\text{len}(I_k) < \delta$, the entire interval $I_k \subseteq (c-\delta, c+\delta)$. So in particular its left endpoint $a_k \in (c-\delta, c+\delta)$. By the construction of each $I_k$, at its left endpoint $f(a_k) < 0$, a contradiction.

If $f(c) < 0$ then let $\varepsilon = -f(c) > 0$. By an exactly symmetric argument there's a neighborhood of $c$ where $f$ is always negative but an interval that fits inside at whose right endpoint $f(b_k) \geq 0$, a contradiction.

Therefore $f(c) = 0$. Since $c \in [a, b]$ but $f(a) < 0 < f(b)$ we get $c \neq a$ and $c \neq b$. Therefore $c \in (a, b)$. ■

## The rest of the theorem

**Theorem 11**: Let $f : [a, b] \to \mathbb{R}$ be continuous. If $L$ is a real number satisfying $f(a) < L < f(b)$ or $f(a) > L > f(b)$, then there exists a point $c \in (a, b)$ where $f(c) = L$.

*Proof*: Suppose $f(a) < L < f(b)$. Then define

$$g(x) = f(x) - L.$$

Then $g$ is continuous by the Algebraic Continuity Theorem and satisfies

$$g(a) < 0 < g(b),$$

so by the preceding results there exists some $c \in (a, b)$ such that $g(c) = 0$. Then $f(c) - L = 0$ so $f(c) = L$.

Now suppose that $f(a) > L > f(b)$, so

$$-f(a) < -L < -f(b),$$

so

$$L - f(a) < 0 < L - f(b).$$

Define

$$g(x) = L - f(x)$$

which is continuous by the Algebraic Continuity Theorem. The inequality above shows that $g(a) < 0 < g(b)$, so there exists some $c \in (a, b)$ such that $g(c) = 0$. Then $L - f(c) = 0$ so $f(c) = L$. ■

## Problem 6: A continuous $f : [0, 1] \to [0, 1]$ has a fixed point

(Exercise 4.5.7)

**Theorem 12**: Let $f$ be a continuous function on the closed interval $[0, 1]$ with range also contained in $[0, 1]$. Then $f$ has a fixed point, that is,

$$f(x) = x$$

for at least one value of $x \in [0, 1]$.

*Proof*: Consider

$$g(x) = f(x) - x.$$

By the Algebraic Continuity Theorem, $g(x)$ is continuous on $[0, 1]$.

The range of $f$ is contained in $[0, 1]$, so $0 \leq f(x) \leq 1$ for every $x \in [0, 1]$. Therefore $g(0) = f(0) \geq 0$, while $g(1) = f(1) - 1 \leq 0$. That is,

$$g(0) \geq 0 \geq g(1).$$

If $g(0) = 0$ or $g(1) = 0$ then we are done, for then the required point would be 0 or 1 respectively. Otherwise we may assume

$$g(0) > 0 > g(1).$$

By the Intermediate Value Theorem, there exists a point $c \in (0, 1)$ such that $g(c) = 0$, i.e., $f(c) = c$. ∎